

Section 96 — Master Continuous Geometry Mixed Contest

Contest contract

Six original problems revisit the predicates and optimization structures of Sections 94–95 under different models.

Problem	Main tool
A	half-plane intersection and shoelace area
B	inward offsets plus feasibility search
C	circle-overlap classification
D	randomized minimum enclosing circle
E	Delaunay empty circumcircles
F	Delaunay sparsification plus Kruskal

For every floating solution, state the exact combinatorial event being approximated. Epsilon is a policy for a known predicate, not a substitute for choosing the predicate.

Recognition and proof map

- Intersecting oriented linear constraints calls for half-plane intersection.
- Maximizing a radius inside a convex region becomes feasibility of inward offsets, followed by monotone search.
- Two-circle formulas are used only after disjointness and containment are classified.
- A smallest covering disk is fixed by at most three boundary witnesses.
- Empty circumcircles point toward Delaunay/Voronoi structure.
- A complete geometric graph whose desired edges have empty-region certificates may be safely sparsified before Kruskal.

For inward offsets, moving each counterclockwise polygon edge left by r describes exactly the possible centers at distance at least r from that edge. If radius r is feasible, every smaller radius is feasible, so binary search is justified. The final numerical tolerance should be derived from the desired answer error and number of iterations, separately from predicate tolerances.

Metamorphic tests

Translate, rotate, or uniformly scale an instance and check that incidence is unchanged while lengths and areas scale correctly. Permute input points to test randomized incremental algorithms. Compare sparse MST with complete MST, half-plane output with clipping on tiny bounded cases, and circle formulas with Monte Carlo only as a rough diagnostic, never as the correctness oracle.

Exercises

A–F form the complete mixed contest.